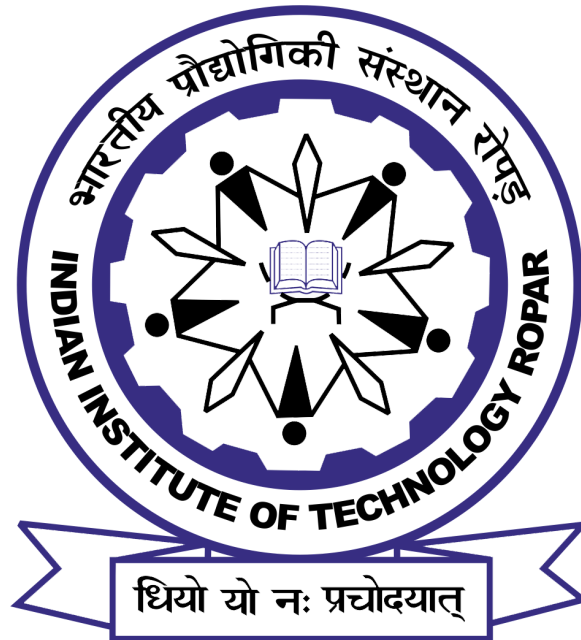


B.Tech Project CP-303

Virtual Instrumentation of Arc Welding

Mid-Semester Report
Department of Mechanical Engineering, IIT Ropar



Supervised by:
Dr. Chandrakant K. Nirala

Submitted by:
Chahat - 2022MEB1046
Arnav Maitreya - 2022MEB1299

Submitted on: 4th March, 2026

Content	2
Chapter 1: Introduction	4
1.1 Background and Motivation	4
1.2 Problem Description	4
1.3 Objectives	5
Chapter 2 : Methodology 1 – Evaluation Techniques	6
2.1 Overview	6
2.2 Pearson Correlation Coefficient	6
3.3 Mean Squared Error (MSE)	6
3.4 Dynamic Time Warping (DTW)	7
3.5 Statistical Feature Comparison	7
Chapter 4 : Methodology 2 – Synthetic Signal Generator	8
4.1 Signal Segmentation	8
The first step in synthetic signal generation was segmenting the welding signal into individual peaks.	8
Since welding signals are continuous time-series signals, segmentation was required to isolate individual welding events.	8
The segmentation process involved:	8
• Detecting peaks in current or power signals	8
• Identifying the start and end points of each peak	8
• Extracting a time window around each peak	8
Each segmented peak represents a single welding arc event, which forms the basis for synthetic signal generation.	8
4.2 Generative Adversarial Networks (GAN)	8
4.3 Spring-Mass Displacement Calculation	9
4.4 Markov Model	9
4.5 Semi-Markov Model	9
Chapter 5 : Work Progress	10
5.1 Data Collection	10
5.2 Signal Preprocessing	10
5.3 Feature Extraction	11
5.4 Synthetic Signal Generation Experiments	11
5.5 Development of Semi-Markov Model	11
Chapter 6: Challenges Faced	12
6.1 Limited Dataset Availability	12
This limitation significantly affected the performance of generative models.	12
6.2 Peak Alignment Issues	12
6.3 Autocorrelation Function (ACF) Mismatch	13
6.4 Incorrect Signal Distribution	13
6.5 Sensitivity to Input Data	13
6.7 Instability in GAN Training	14
Chapter 7: Results - Development of Dashboard for Result Visualization	15

7.1 Visualization of Welding Signals	16
7.2 Peak Analysis	16
7.3 Correlation and Evaluation Metrics	16
7.4 Interactive Analysis	16
7.5 Purpose of the Dashboard	16
Chapter 9: Conclusion	17
Chapter 10: References	1

Chapter 1: Introduction

1.1 Background and Motivation

Welding is one of the most widely used manufacturing processes in industries such as automotive, aerospace, construction, and heavy engineering. The quality and strength of welded joints depend heavily on electrical parameters like Voltage (V) and Current (I) during the welding process. These parameters vary continuously with time and reflect the behavior of the welding arc.

Modern welding systems allow monitoring of electrical signals through Virtual Instrumentation, which records time-series data of voltage and current using computer-based measurement systems. By analyzing these signals, it is possible to understand welding behavior, detect abnormalities, and monitor weld quality.

However, raw welding signals are continuous and complex, making it difficult to directly interpret welding conditions. Therefore, signal processing and data analytics techniques are required to extract useful information from these signals.

This project focuses on analyzing welding electrical signals, identifying individual welding events (peaks), and categorizing them into meaningful classes such as normal arcing, effective welding, and short circuits.

1.2 Problem Description

Modern welding machines generate large amounts of electrical signal data such as Voltage (V), Current (I), and Power (P) with respect to time. These signals contain valuable information about welding arc behavior and weld quality.

Traditionally, to obtain sufficient datasets for analysis, welding machines must be operated repeatedly under different parameter settings. This requires the machine to run continuously for long durations, which leads to:

- High electrical energy consumption
- Increased equipment wear
- Higher operational cost
- Time-consuming experimentation

Additionally, the signals generated during welding are continuous and complex, making it difficult to identify individual welding events and categorize them into different welding conditions. Therefore, there is a need to develop a system that can:

- Analyze welding voltage and current signals

- Detect and segment welding peaks
- Categorize welding events into meaningful classes
- Generate additional realistic datasets without repeatedly running the welding machine

By generating synthetic or augmented datasets from existing welding signal data, the proposed system can reduce the need to operate welding machines solely for data collection, thereby saving a large amount of electrical energy and reducing experimental costs.

1.3 Objectives

The main objectives of the project are:

- To collect welding electrical signal data using Virtual Instrumentation.
- To analyze voltage and current signals obtained from the welding process.
- To preprocess and clean the signal data for accurate analysis.
- To detect peaks in the welding signal representing individual welding events.
- To extract meaningful features from each welding peak.
- To generate additional realistic datasets from existing signals.
- To classify each welding peak into categories such as:
 - Normal arcing
 - Effective welding
 - Short circuit
- To develop a structured dataset that can be used for intelligent welding analysis.

Chapter 2 : Methodology 1 – Evaluation Techniques

2.1 Overview

Evaluation techniques were used to assess the similarity between the original welding signals and the generated synthetic signals. These techniques help determine whether the synthetic data preserves the statistical and temporal characteristics of the real welding signals.

The evaluation focused on measuring correlation, error, and similarity between the real and generated signals.

2.2 Pearson Correlation Coefficient

The Pearson correlation coefficient was used to measure the **linear relationship between real and synthetic signals**.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where:

- x_i = real signal values
- y_i = synthetic signal values
- $\bar{x}, \bar{y}$ = mean values

A correlation value close to **1** indicates strong similarity between signals.

3.3 Mean Squared Error (MSE)

Mean Squared Error measures the **average squared difference between real and synthetic signals**.

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2$$

Where:

- x_i = real signal value
- y_i = synthetic signal value
- n = number of samples

Lower MSE values indicate that the generated signals are closer to the real signals.

3.4 Dynamic Time Warping (DTW)

Dynamic Time Warping (DTW) measures similarity between **two time-series signals that may vary in speed or duration**.

This metric is useful because welding signals may have peaks that occur at slightly different time intervals while still representing the same welding behavior.

3.5 Statistical Feature Comparison

Statistical properties of real and synthetic signals were also compared.

The following features were analyzed:

- Mean value
- Variance
- Standard deviation
- Peak amplitude
- Signal duration

This comparison ensured that the generated signals maintained the **statistical characteristics of the original dataset**.

Chapter 4 : Methodology 2 – Synthetic Signal Generator

4.1 Signal Segmentation

The first step in synthetic signal generation was segmenting the welding signal into individual peaks.

Since welding signals are continuous time-series signals, segmentation was required to isolate individual welding events.

The segmentation process involved:

- Detecting peaks in current or power signals
- Identifying the start and end points of each peak
- Extracting a time window around each peak

Each segmented peak represents a single welding arc event, which forms the basis for synthetic signal generation.

4.2 Generative Adversarial Networks (GAN)

The initial approach used for generating synthetic welding signals was **Generative Adversarial Networks (GANs)**.

A GAN consists of two neural networks:

Generator

The generator produces synthetic welding signals that resemble the real signals.

Discriminator

The discriminator evaluates whether the generated signal is real or synthetic.

The GAN objective function is:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

However, GAN models did not perform well due to:

- Limited dataset size
- Training instability
- Difficulty capturing complex welding signal patterns

4.3 Spring-Mass Displacement Calculation

To better handle time-series data, **TimeGAN** was explored.

TimeGAN integrates:

- Generative Adversarial Networks
- Recurrent neural networks
- Temporal embedding

It aims to preserve both:

- Feature distribution
- Temporal dynamics

Despite these advantages, the generated signals did not closely match the real welding signals.

4.4 Markov Model

A Markov model was then explored to represent welding signal transitions.

The Markov assumption states:

$$P(X_{n+1}=j | X_n=i)$$

Meaning that the next state depends only on the current state.

However, Markov models assume **constant time duration in each state**, which does not accurately represent welding signals where state durations vary.

4.5 Semi-Markov Model

To overcome the limitations of Markov models, a **Semi-Markov model** was implemented.

A Semi-Markov process allows transitions between states with **variable time durations**.

The transition probability is defined as:

$$P_{ij}(t)=P(X_{n+1}=j, T_{n+1}-T_n \leq t | X_n=i)$$

Where:

- X_n = current state
- X_{n+1} = next state
- T_n = time of transition

The model includes:

- State transition matrix
- Sojourn time distribution

This approach provided a better representation of the temporal behavior of welding signals and produced more realistic synthetic data.

Chapter 5 : Work Progress

The project progressed through multiple stages, starting from data acquisition and signal analysis to synthetic signal generation and evaluation.

5.1 Data Collection

The first stage involved collecting welding electrical signals using a **virtual instrumentation system**. Voltage and current signals generated during the welding process were recorded as time-series data. These signals represent the behavior of the welding arc under different welding conditions.

The collected data was stored in tabular format and visualized as voltage-time and current-time graphs.

5.2 Signal Preprocessing

The raw welding signals contained noise and fluctuations due to electrical disturbances and sensor measurements. Therefore, preprocessing techniques were applied to improve signal quality.

This stage involved:

- Noise filtering
- Signal smoothing
- Normalization of voltage and current values

These steps helped obtain clean signals suitable for further analysis.

5.3 Feature Extraction

After segmentation, different features were extracted from each welding peak to describe the signal behavior numerically.

The extracted features included:

- Maximum current
- Minimum voltage
- Peak power
- Peak duration
- Rise time and fall time
- Statistical measures such as mean and variance

These features helped represent the welding signals in a structured format for further modeling.

5.4 Synthetic Signal Generation Experiments

To expand the dataset, several synthetic signal generation techniques were explored.

Initially, **Generative Adversarial Networks (GAN)** were implemented to generate artificial welding signals. However, the generated signals did not accurately represent the real welding patterns.

Next, **TimeGAN**, a model designed specifically for time-series data, was implemented. Although it improved temporal modeling, the generated signals still showed mismatches with the original welding data.

5.5 Development of Semi-Markov Model

A Markov model was first tested to represent transitions between welding states. However, the Markov model assumed fixed state durations, which did not reflect the variability present in welding signals.

To overcome this limitation, a **Semi-Markov model** was implemented. This model allows variable durations in each state and better captures the temporal characteristics of welding signals.

The Semi-Markov model was used to generate synthetic signal sequences that closely resemble the behavior of real welding signals.

Chapter 6: Challenges Faced

During the development of the project, several technical challenges were encountered while analyzing welding signals and generating synthetic data. These challenges were mainly related to signal characteristics, modeling limitations, and evaluation of the generated signals.

6.1 Limited Dataset Availability

One of the major challenges faced during the project was the limited availability of welding signal data. Deep learning models such as GAN and TimeGAN typically require large datasets to learn complex signal patterns effectively.

However, the available dataset contained a limited number of welding signal samples. As a result:

- The models could not capture the full variability of welding signals.
- Training instability occurred in deep learning models.
- Generated synthetic signals lacked diversity.

This limitation significantly affected the performance of generative models.

6.2 Peak Alignment Issues

Another important challenge was related to peak alignment in synthetic signals.

In the original welding signals, peaks correspond to welding arc events and occur at specific intervals. However, when synthetic signals were generated:

- The peaks did not occur at the correct time locations.
- The number of peaks per unit time differed from the real signals.
- Some generated signals contained extra peaks, while others had missing peaks.

This mismatch made it difficult to accurately reproduce the temporal characteristics of the original welding signals.

6.3 Autocorrelation Function (ACF) Mismatch

The Autocorrelation Function (ACF) was used to analyze the temporal dependency within the welding signals.

When comparing the ACF of the real signals and synthetic signals, it was observed that:

- The autocorrelation patterns did not match closely.
- The generated signals failed to preserve the temporal relationships present in the original dataset.
- Periodic characteristics of the welding signal were not accurately captured.

This indicated that the synthetic models were not fully learning the underlying temporal structure of the welding process.

6.4 Incorrect Signal Distribution

Another issue encountered was that the statistical distribution of the generated signals differed from the original signals.

For example:

- Mean values of voltage or current sometimes deviated significantly.
- Variance of generated signals was either too high or too low.
- Peak amplitudes were not consistent with real welding signals.

These discrepancies indicated that the models were not properly capturing the statistical characteristics of the data.

6.5 Sensitivity to Input Data

The models used for synthetic signal generation were highly sensitive to the quality and structure of the input data.

Challenges related to input data included:

- Missing or inconsistent data points
- Noise in the raw signals
- Improper segmentation of peaks

If the input signals were not properly preprocessed or segmented, the generated synthetic signals became inaccurate.

6.6 Difficulty in Modeling Temporal Variability

Welding signals exhibit significant temporal variability, meaning that:

- The duration of welding peaks varies.
- The time between peaks is not constant.
- Signal amplitudes change depending on welding parameters.

Traditional models such as Markov chains assume fixed transition timing, which made it difficult to represent the real welding signal behavior accurately.

This limitation motivated the use of Semi-Markov models, which allow variable state durations.

6.7 Instability in GAN Training

During the implementation of GAN-based models, several training issues were observed:

- Mode collapse, where the generator produces similar signals repeatedly.
- Difficulty in balancing the generator and discriminator networks.
- Slow convergence during training.

These issues resulted in synthetic signals that did not accurately represent the real welding signal patterns.

6.8 Model Selection Challenges

Selecting an appropriate model for synthetic signal generation required experimentation with multiple approaches.

Several models were explored:

- GAN
- TimeGAN
- Markov Model
- Semi-Markov Model

Each model had advantages and limitations. Identifying the most suitable model required extensive testing and evaluation.

Eventually, the **Semi-Markov model** was selected because it could better capture the temporal behavior and state duration variability present in welding signals.

6.9 Evaluation Difficulties

Evaluating the quality of synthetic signals was also challenging. Multiple metrics had to be used to determine how closely the generated signals matched the real signals.

These included:

- Pearson correlation
- Mean Squared Error (MSE)
- Dynamic Time Warping (DTW)
- Autocorrelation comparison

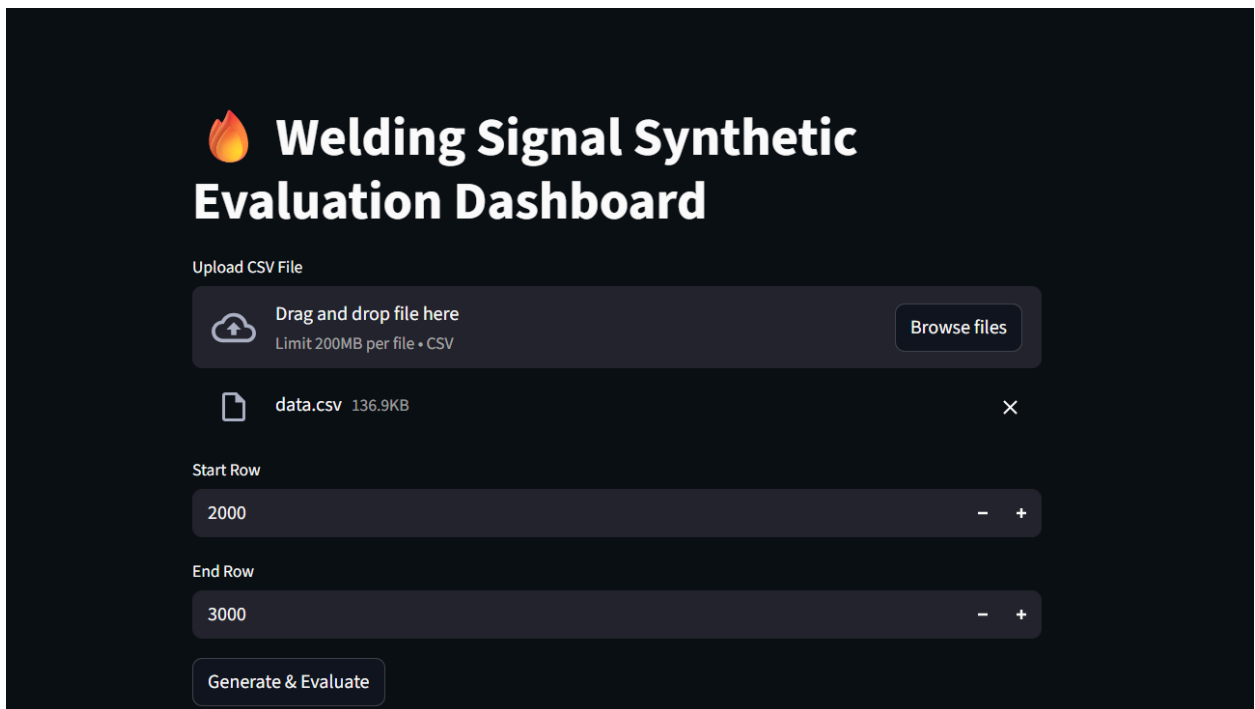
Interpreting these metrics together to determine the effectiveness of the models required careful analysis.

Chapter 7: Results - Development of Dashboard for Result Visualization

After implementing the **Semi-Markov model** and generating synthetic welding signals, a dashboard was developed to visualize and analyze the results in an interactive manner. The purpose of the dashboard was to provide a clear and organized way to compare the **real welding signals and the generated synthetic signals**.

The dashboard integrates different visualization components to display important aspects of the analysis, including signal behavior, peak distribution, and statistical comparison.

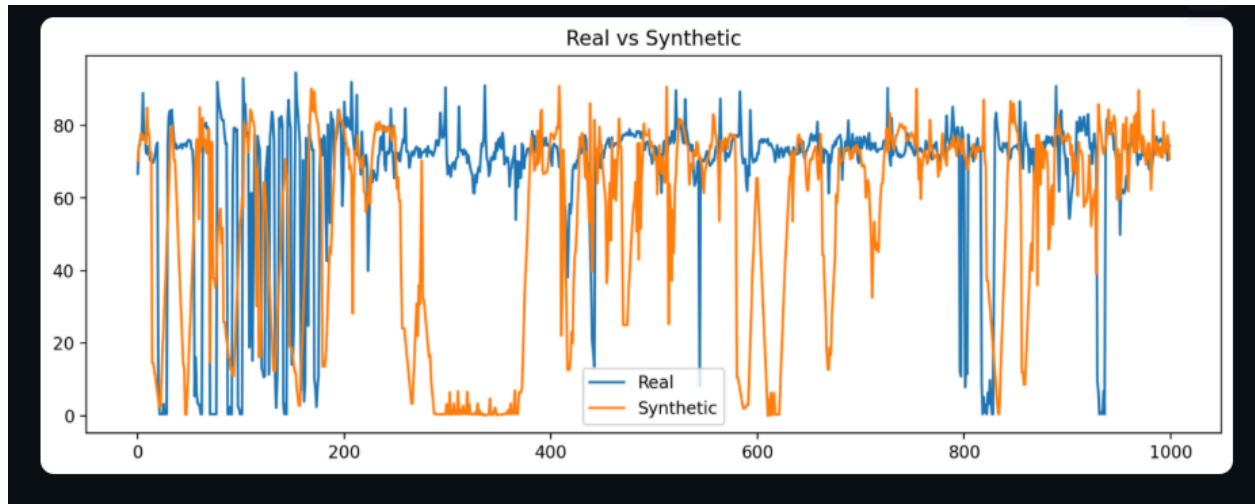
The dashboard includes the following features:



The screenshot displays the 'Welding Signal Synthetic Evaluation Dashboard'. At the top, there is a title 'Welding Signal Synthetic Evaluation Dashboard' with a flame icon. Below the title, there is a section for 'Upload CSV File'. This section includes a drag-and-drop area with the text 'Drag and drop file here' and 'Limit 200MB per file • CSV', a 'Browse files' button, and a file upload indicator showing 'data.csv' with a size of '136.9KB' and a close button. Below the upload section, there are two input fields for 'Start Row' (set to 2000) and 'End Row' (set to 3000), each with minus and plus buttons for adjustment. At the bottom, there is a 'Generate & Evaluate' button.

7.1 Visualization of Welding Signals

Graphs were created to display **voltage vs time** and **current vs time** signals for both real and synthetic datasets. This helped in visually verifying whether the generated signals follow similar patterns as the original welding signals.



7.2 Peak Analysis

The dashboard displays detected peaks for each signal segment. This allows comparison of the **number of peaks per unit time**, peak amplitudes, and peak intervals between real and synthetic signals.

Statistical Comparison

Key statistical properties of the signals such as:

- Mean
- Variance
- Standard deviation
- Peak amplitude

were displayed in the dashboard to compare the similarity between real and generated signals.

7.3 Correlation and Evaluation Metrics

The dashboard also presents the calculated evaluation metrics used in the project, including:

- Pearson correlation coefficient
- Mean Squared Error (MSE)
- Dynamic Time Warping (DTW)
- Autocorrelation analysis

These metrics help quantify the similarity between the real and synthetic datasets.



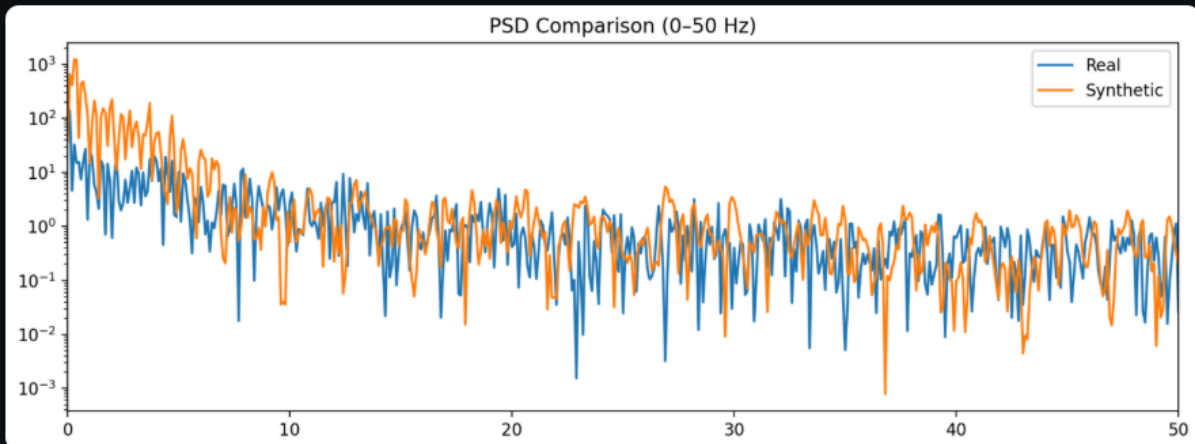
Core Metrics

RMS % Error: 14.654

PSD Peak % Difference: 791.042

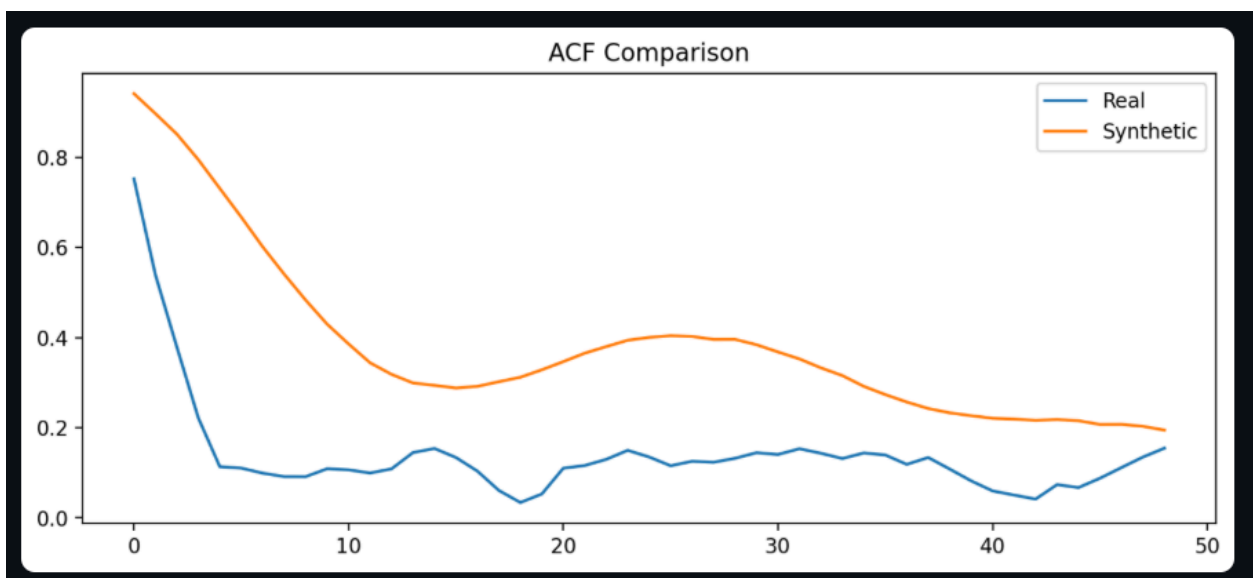
ACF Correlation: 0.721

Kurtosis % Error: 118.073



7.4 Interactive Analysis

The dashboard allows easier interpretation of results by presenting multiple visualizations and metrics in a single interface. This helps quickly evaluate the effectiveness of the **Semi-Markov synthetic signal generator**.



7.5 Purpose of the Dashboard

The dashboard serves as a tool for:

- Visual validation of synthetic signals
- Comparison between real and generated data
- Monitoring evaluation metrics
- Presenting results in an intuitive and organized manner

This visualization platform makes it easier to interpret the performance of the developed synthetic signal generation model.

- **Chapter 8: Future Scope**

8.1 Integration of Audio Signals with Welding Signals

In future work, the audio signals generated during the welding process can be recorded and synchronized with the corresponding voltage and current signals obtained from the welding machine. Welding arcs produce characteristic sound patterns that reflect the behavior of the welding process. By collecting both electrical and audio signals simultaneously, it will be possible to analyze how the sound produced during welding relates to the variations in welding current and arc stability.

Chapter 9: Conclusion

This project focused on the analysis of welding arc signals using **virtual instrumentation and data analytics** techniques. The objective of the work was to analyze voltage and current signals generated during the welding process, segment the signals into individual welding events, and generate synthetic signals to expand the available dataset without repeatedly operating the welding machine.

Initially, welding signal data was collected and processed to remove noise and irregularities. The signals were then segmented into individual peaks corresponding to welding arc events. From these segmented peaks, several electrical, temporal, and statistical features were extracted to better understand the behavior of the welding process.

To address the challenge of limited datasets, multiple synthetic signal generation approaches were explored. Generative models such as **GAN and TimeGAN** were initially implemented to generate time-series signals. However, due to limited data availability and the complex temporal nature of welding signals, these models did not produce satisfactory results.

Subsequently, probabilistic approaches were explored. A **Markov model** was tested but was found to be limited because it assumes constant state durations. To overcome this limitation, a **Semi-Markov model** was implemented, which allows variable state durations and better represents the temporal characteristics of welding signals.

The synthetic signals generated using the Semi-Markov model were evaluated using several statistical and correlation metrics, including **Pearson correlation, Mean Squared Error (MSE), Dynamic Time Warping (DTW), and autocorrelation analysis**. The results showed that the semi-Markov approach produced synthetic signals that preserved the statistical and temporal characteristics of the original welding signals more effectively than the previously tested models.

One of the key contributions of this work is the ability to generate realistic welding signal data without repeatedly running the welding machine. This approach helps **reduce energy consumption, experimental cost, and machine usage**, making the process more efficient and sustainable.

Overall, the project demonstrates that **probabilistic temporal models such as Semi-Markov models can effectively generate synthetic welding signals and support further analysis of welding processes**. The developed methodology can be used as a foundation for future work in welding process monitoring, intelligent welding systems, and machine learning-based quality prediction.

